

Improvements on NRAO Proposal Classification Model

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Background



About NRAO

What is the NRAO?

U.S. national facility for radio astronomy, funded by the National Science Foundation

What does the NRAO do?

Manages world-class observatories including ALMA, the VLA, and Green Bank, supporting global astronomical research

Where is the NRAO HQ?

Charlottesville, VA!



National Radio
Astronomy
Observatory

About ALMA

(Atacama Large Millimeter/submillimeter Array)



What is ALMA?

A state-of-the-art radio telescope
located in Chile's Atacama Desert

What does ALMA do?

Observes the Universe in millimeter
and submillimeter wavelengths



Previous Work

- Suggested frequencies from proposal abstracts
- Used LDA+ HDBSCAN + Naive Bayes
- Matched “areas of interest” with 67%+ accuracy

Optimizing the ALMA Research Proposal Process with Machine Learning

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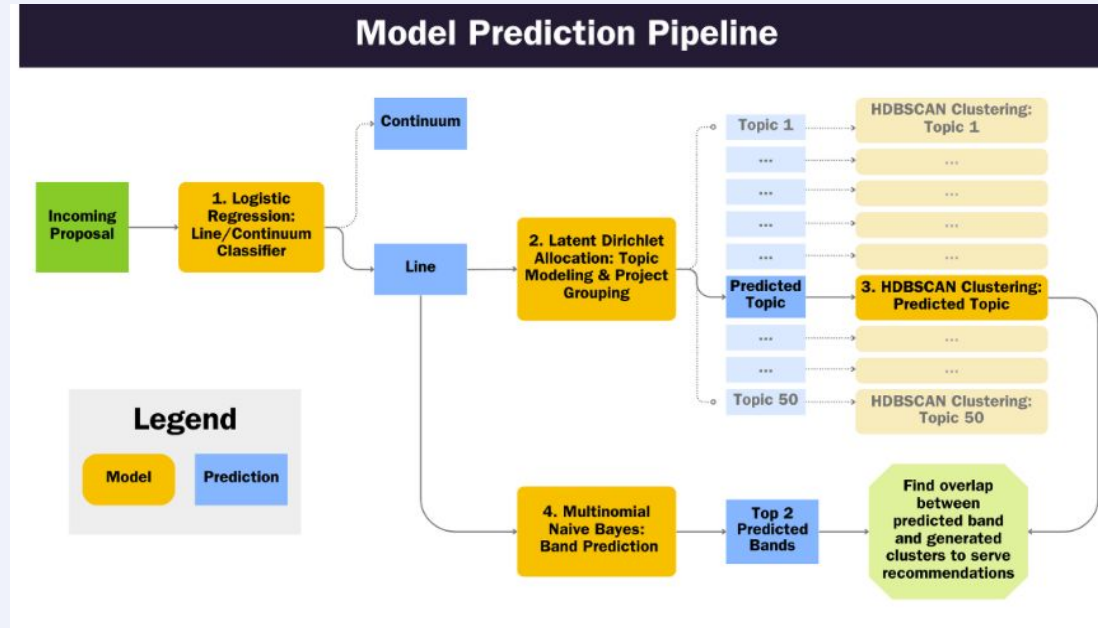
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Abstract— Every year, astronomers from around the world submit research proposals to the Atacama Large Millimeter Array (ALMA), the largest radio telescope array in the world. The aim of the current work is to streamline the proposal process for astronomers submitting projects to ALMA by suggesting frequency ranges that may be relevant to their research based on their proposal text. We introduce a pipeline of supervised and unsupervised machine learning models, each using various representations of the title and abstract of an incoming proposal. First, a logistic regression filters out proposed projects that are not expected to need specific technical setups. Second, if a technical setup is deemed necessary, our pipeline assigns an incoming project to one of 50 “similar project” groups, defined by topics generated from Latent Dirichlet Allocation (LDA). Third, we apply Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) to mine patterns in

the telescope array allows ALMA to detect electromagnetic radiation on a fairly continuous range of frequencies from 35 GHz to 950 GHz at two-decimal-point precision. This continuous range of frequencies is broken up into ten discrete bands. Furthermore, the technical specifications of the telescope allow individual measurements to span a range of approximately 4 GHz. On top of the technical nature of project proposals, ALMA can only accept around 400 proposals to enact throughout the year, making it a valuable and competitive resource. Thus, it is paramount that researchers submit well designed and technically proficient proposals to increase their chances of acceptance.

Our work streamlines the ALMA proposal process by providing researchers insight and assistance on how to set their

Previous Work



Takeaways from Previous Work

- Frequency prediction is feasible and helpful
- Clustering + classification enhances model precision
- Opportunities exist to go deeper with NLP models and AstroBERT

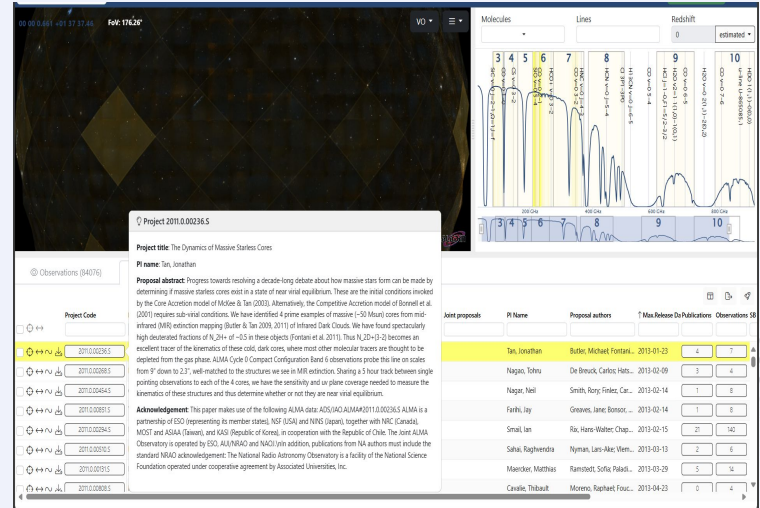


About the Data



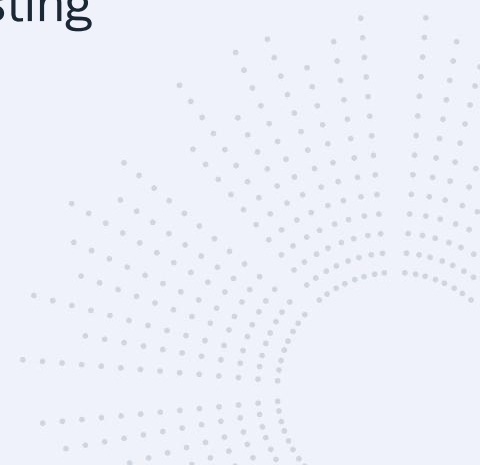
Project Proposals

- Line vs. Continuum
- Filtered out projects with >2 bands
- 3178 Project Proposals
- 23,482 total measurements used for modeling



Limitations

- Focused only on "line" setups, not continuum projects
- Unable to obtain full proposal text due to concern over training models on researchers' material
- Our frequency recommendations are based on existing measurements



Methodology



Topic Modelling

- Comparison
- astroBERT
 - Astrophysics-based language model
 - Designed for semantic understanding of ADS projects
 - Additional level of clustering required for topic generation
- Performance Metric: **Average Measurement**

Hit Rate per Project



Analysis Results



Topic Modelling (LDA)

<u>Topic Generation</u>	<u>Clustering</u>	<u>Average Hit Rate per Project</u>
<i>Latent Dirichlet Allocation</i>	<i>HDBSCAN (previous work)</i>	88%
Latent Dirichlet Allocation	OPTICS	86%
Latent Dirichlet Allocation	Spectral Clustering	96%

Topic Modelling (astroBERT)

<u>Topic Generation</u>	<u>Clustering</u>	<u>Average Hit Rate per Project</u>
AstroBERT + HDBSCAN	Spectral Clustering	49%
AstroBERT + Spectral Clustering	Spectral Clustering	43%
AstroBERT + KMeans	Spectral Clustering	86%
AstroBERT + Gaussian Mixture	Spectral Clustering	87%
AstroBERT + Dirichlet Process	Spectral Clustering	88%

Challenges

- Density-based clustering not conducive to inference
 - Used distance to centroid as a proxy
- Clustering $\neq$ Classification
- Topic Saliency
 - Generic Keywords (e.g. ‘observation’)
 - Removing additional stopwords resulted in insignificant performance difference



Dashboard



Dashboard Details

- Topic Selector
- Band Selector
- Cluster Distribution
- Most Common Chemical
- Demo

Calculate the Most Common Chemicals

Lower Bound (K): Upper Bound (K):

Below is a table that calculates how frequent a certain chemical shows up in the selected cluster frequency ranges, shown by the count variable.

LSE stands for lower state energy, measured in Kelvin, and split up by quartiles.

Most Common Chemicals

chemical_name	count	LSE_Min	LSE_Q25	LSE_Q50	LSE_Q75
ethyl cyanide	4039	15.39	262.85	576.76	1014.84
vinyl cyanide	3366	18.15	538.57	789.18	1192.83
methyl formate	2253	14.7	307.9	498.36	729.31
acetone	1857	38.35	287.57	831.03	1456.72
acetic acid	1658	21.74	265.4	564.34	1149.45
acetamide	1421	24.48	146.33	193.91	251.99
ethylene glycol	1124	45.69	245.35	456.58	686.46
acetaldehyde	1121	10.43	310.26	476.13	841.03
iso-propyl cyanide	986	51.77	337.06	563.87	925.87
aluminum monochloride	885	132.87	132.87	132.87	132.87

Moving Forward



Moving Forward

- Tested many different topic generation/clustering techniques
- In depth importance on specific molecule lines
 - Domain Knowledge
- LLM Project



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